

```

const int trig=10;
const int echo=9;
float distance;
long duration;
int IN4=2;
int IN3=3;
int EN3=6;
int r_led=4;
int g_led=5;

void setup()
{
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  pinMode(EN3, OUTPUT);
  pinMode(r_led, OUTPUT);
  pinMode(g_led, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  digitalWrite(trig, LOW);
  delayMicroseconds(2);

  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig, LOW);

  duration=pulseIn(echo, HIGH);
  distance= duration*0.0343/2;
  delay(500);
  if (distance>=20 && distance<=60)
  {
    int brightness= map (distance, 20, 60, 0, 255);
    analogWrite(r_led, brightness);
    analogWrite(g_led, LOW);
    analogWrite(IN3, HIGH);
    analogWrite(IN4, LOW);
    analogWrite(EN3, 50);
    Serial.println(brightness);}
  else if (distance>60 && distance<200)
  {
    int brightness= map (distance, 60, 200, 0, 255);
    analogWrite(g_led, brightness);
    analogWrite(r_led, LOW);
    analogWrite(IN3, HIGH);
    analogWrite(IN4, LOW);
    analogWrite(EN3, 100);
    Serial.println(brightness);}
  else
  {
    analogWrite(r_led, LOW);
    analogWrite(g_led, LOW);
    analogWrite(IN3, HIGH);
    analogWrite(IN4, LOW);
    analogWrite(EN3, 0);}
}

```

```
    Serial.print(distance);  
}
```